



KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year II Semester Regular Examinations, June/July-2025

DATA STRUCTURES

(COMPUTER SCIENCE AND ENGINEERING)

Time: 3 hours

Max. Marks: 60

Note: This question paper contains two parts A and B.

Part A is compulsory which carries 10 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b as sub questions.

PART- A

(10 Marks)

1. a) How are templates useful in C++. [1M] L2
- b) Define Polymorphism. [1M] L1
- c) Define a sparse matrix with an example. [1M] L1
- d) What is indirect recursion? [1M] L2
- e) Define push and pop operations with respect to stack. [1M] L1
- f) Write a C++ code snippet to insert element into queue. [1M] L2
- g) What are the different types of tree traversal methods. [1M] L1
- h) What is the maximum number of nodes in a binary tree of height 'h'? [1M] L2
- i) Define a priority queue. [1M] L1
- j) What is the difference between adjacency matrix and adjacency list of a graph? [1M] L2

PART- B

(5 X 10 =50 Marks)

2. a) Write a C++ program using a base class Shape with a virtual function area() and derive classes Rectangle and Circle that override the function. [5M] L3
- b) What are the default arguments in C++ functions? Show how they work with a program. [5M] L3

OR

3. a) What happens when you do not define a destructor in your class? Explain with an example program where the class allocates dynamic memory. [5M] L2
- b) What is the difference between local and global variables in terms of memory allocation and accessibility? Explain with a program. [5M] L2
4. a) What is recursion? Explain with an example by writing a recursive function to calculate the factorial of given number. [5M] L2
- b) Write a C++ program to insert a node at the beginning and display the list using Single Linked List. [5M] L3

OR

5. a) Distinguish between dynamic arrays and static arrays. [4M] L2
b) What is forward traversal in a Doubly Linked List. Explain with the help of program and example. [6M] L3
6. a) You are developing a simple text editor that supports undo operations. Explain how you can use the stack to implement the undo operations. Write a C++ program that pushes operation on to a stack and performs undo. [6M] L3
b) Convert following arithmetic infix expression into postfix by using stack: [4M] L3
 $a*(b+c)+(b/d)*a+z*u$

OR

7. a) A music streaming app buffers upcoming songs to ensure smooth playback. How can a queue be used to manage preloaded songs? Describe how songs are added to and removed from the queue as playback continues with the help of a program. [5M] L3
b) Write a C++ program to implement a Queue using arrays. Include enqueue and display operations. [5M] L3
8. a) Construct a binary search tree for the given values 50, 30, 70, 20, 40, 60, 80 and draw the final BST. [6M] L3
b) Explain the properties of a Binary Tree. Support your answer with definitions and relevant examples. [4M] L3

OR

9. a) Write a C++ program to perform post order traversal of a binary tree. [5M] L4
b) Trace the BFS traversal for the following graph (represented as an adjacency list):
Graph:
A → B, C
B → D
C → E
D → A
E → D [5M] L4
10. What is hash function and explain different types of hash functions with examples. [10M] L2

OR

11. Write a C++ program for BFS traversal methods in a graph with an example. [10M] L2

---ooOoo---



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B.Tech II Year II Semester Regular Examinations, June/July-2025

DATA STRUCTURES

(COMMON TO IT, CSE(AI&ML), CSE(DS))

Time: 3 hours

Max. Marks: 60

Note: This question paper contains two parts A and B.

Part A is compulsory which carries 10 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b as sub questions.

PART- A

(10 Marks)

1. a) List the different types of templates in C++.

[1M] L1

b) What is a default argument in a function?

[1M] L1

c) What is direct recursion?

[1M] L1

d) Name different types of Linked List.

[1M] L1

e) Convert the infix expression $A+B*D/E+F-G*H$ into prefix expression.

[1M] L2

f) Define enqueue() and dequeue() operations.

[1M] L1

g) Define full binary tree and complete binary tree.

[1M] L2

h) Define root, leaf and internal nodes.

[1M] L1

i) What is Connected Graph?

[1M] L1

j) What is linear probing in collision resolution?

[1M] L1

PART- B

(5 X 10 =50 Marks)

2. a) Define a class. Explain different types of access specifiers with an example program.

[5M] L2

b) Write a function that swaps two numbers using pass-by-value and pass-by-reference. Explain how the results differ.

[5M] L2

OR

3. a) What is the purpose of 'this' pointer in C++? Provide a code snippet to illustrate its usage.

[5M] L2

b) Write a class Student that has both default, parameterized, and copy constructors. Show how each is invoked.

[5M] L3

4. a) Define a Doubly Linked List. How is it different from a singly linked list?

[2M] L2

b) Define Sparse matrix. Write the working of sparse matrix using array implementation.

[8M] L3

OR

5. a) Write a function to search for a given element in a Singly Linked List. [5M] L2
b) Write a C++ program to compute the sum of all elements in a 2D array. [5M] L2
6. Explain differences between stack and queue with examples. Write a C++ program to find the minimum element in the stack using Linked List. [10M] L2
- OR
7. Write a C++ program to implement Stack using Linked List with appropriate Abstract Data Type. [10M] L2
8. a) Write a C++ program to implement tree traversal methods with suitable examples [5M] L4
b) Construct a binary search tree for the given values 25, 15, 50, 10, 22, 35, 70 and draw the final BST. [5M] L3
- OR
9. a) Write a C++ program to perform pre order traversal of a binary tree. [5M] L4
b) Construct a binary tree using the input: 1, 2, 3, 4, 5, NULL, 6 (level-order input). Write the BFS traversal of the tree. [5M] L4
10. What is Collision in Hashing. Explain different types of Collision resolution techniques with examples. [10M] L3
- OR
11. Write a C++ program for DFS traversal methods in a Graph with an example. [10M] L3

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Code No: 223AC

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KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

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Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year I Semester Supplementary Examinations, July-2025

DATA STRUCTURES THROUGH C++

(COMMON TO CSE, IT, CSE(AI&ML), CSE(DS))

Time: 3 hours

Max. Marks: 70

Note: This question paper contains two parts A and B.

Part A is compulsory which carries 20 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b as sub questions.

PART- A

(20 Marks)

1. a) Define class and object. [2M] L1
- b) Explain the linear and non-linear data structures. [2M] L2
- c) What is queue ADT? [2M] L1
- d) Explain about dynamic array. [2M] L2
- e) What are the operations performed on stack? [2M] L1
- f) Explain the postfix expression [2M] L2
- g) Define a max heap. [2M] L1
- h) What is hash function? [2M] L1
- i) Explain about time complexity of merge sort. [2M] L2
- j) List the graph search methods. [2M] L1

PART- B

(5 X 10 =50 Marks)

2. a) Explain constructors and destructors with examples. [5M] L2
- b) What is an exception? Discuss about throwing an exception and handling an exception. [5M] L2

OR

3. a) Explain new and delete operators with an example programs. [5M] L2
- b) What is recursion? Explain about data abstraction [5M] L2
4. a) Discuss about sparse matrices with examples. [5M] L2
- b) Explain array representation of stack. [5M] L2

OR

5. a) Briefly explain about applications of stack. [5M] L2
b) Explain the Insert and Delete operations in Doubly Linked List data structure. [5M] L2
6. a) Discuss about array implementation of queue ADT [5M] L2
b) Explain the array representation of a threaded binary tree. [5M] L2
- OR
7. a) How to evaluate postfix expression? Explain [5M] L2
b) Discuss the circular lists and header nodes. [5M] L2
8. a) Explain about Insertion and deletion operations in max heap. [5M] L2
b) Distinguish between full binary tree and complete binary tree. [5M] L4
- OR
9. a) Compare array and linked list representation of binary trees. [5M] L2
b) Discuss about the ADT Binary Tree. [5M] L2
10. a) Write a C++ program for heap sort technique. [5M] L3
b) Discuss in detail about red -black trees. [5M] L2
- OR
11. a) Write a C++ program for insertion sort technique [5M] L3
b) Explain the DFS and BFS methods. [5M] L2

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Code No: 213AC

KR21

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B.Tech II Year I Semester Supplementary Examinations, July-2025

DATA STRUCTURES THROUGH C++

(COMMON TO CSE, IT, CSE (AI&ML), CSE(DS))

Time: 3 hours

Max. Marks: 70

Note: This question paper contains two parts A and B.

Part A is compulsory which carries 20 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b as sub questions.

PART- A

(20 Marks)

1. a) What is meant by pass by value parameters? [2M] L1
- b) Show the function overloading in C++. [2M] L2
- c) Write a C++ program to reverse a linked list by traversing it only once. [2M] L1
- d) Explain about queue operations in brief. [2M] L2
- e) Define threaded binary tree. [2M] L1
- f) Outline the procedure to convert a general tree into binary tree. [2M] L2
- g) Compare linear search and binary search. [2M] L2
- h) Show the worst-case time complexity of quick sort. [2M] L1
- i) Define Graph with examples. [2M] L1
- j) Model the Knuth-Morris-Pratt algorithm. [2M] L1

PART- B

(5 X 10 =50 Marks)

2. a) Write a C++ program to swap two numbers using function templates. [5M] L3
- b) Illustrate clearly about priority queues. [5M] L2

OR

3. a) What is Constructor? Explain various types of constructors with an example. [5M] L1
- b) Define inheritance and illustrate clearly about their types. [5M] L1
4. a) What is polymorphism? Explain about the types of polymorphism. [5M] L2
- b) Write a C++ Program to push an element into a stack. [5M] L3

OR

5. a) What are the various applications of stacks? Explain infix to postfix conversion. [5M] L1
b) Discuss about linked list implementation of queue ADT. [5M] L2
6. a) Create binary search tree for the following elements (23, 32, 24, 36, 15, 12, 39, 2, 19) and discuss the height of the above binary search tree and list the properties of BST. [10M] L3
- OR
7. Explain in detail about binary tree traversal and its various traversal techniques. [10M] L2
8. a) Explain about heap sort technique with an example. [5M] L2
b) Construct max heap for the following elements (28, 16, 14, 103, 52, 105, 139, 27, 190). [5M] L3
- OR
9. a) Write a C++ Program to search an element using binary search and trace the above program to search 23 in the following elements 12, 15, 18, 20, 22, 36, 39, 40, 46 which is unsuccessful search. [10M] L3
10. a) Make use of following: a) Applications of graphs b) Spanning tree. [5M] L3
b) Discuss about different ways of representing Graphs in memory. [5M] L2
- OR
11. a) Explain in detail about graph ADT. [5M] L2
b) Explain about adjacency matrix and adjacency list. [5M] L2

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KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

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B.Tech II Year I Semester Supplementary Examinations, February-2024

Data Structures through C++

(Common to CSE, IT, CSE (AI&ML), CSE (DATA SCIENCE))

Time: 3 hours

Max. Marks: 70

Note: This question paper contains two parts A and B.

Part A is compulsory which carries 20 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(2 X 10 =20 Marks)

1. a) What is data structure? List various types of data structure. [2M]L1
- b) What is the use of new and delete operators in C++. [2M]L2
- c) Define Polymorphism. [2M]L1
- d) Write the syntax how to create a node in single link list. [2M]L2
- e) List out the properties of Binary trees. [2M]L1
- f) What is the height of a tree? Give an example. [2M]L2
- g) Define Hash function. [2M]L1
- h) Compare different sorting techniques. [2M]L2
- i) List the applications of graphs. [2M]L1
- j) What is minimum cost spanning tree? [2M]L1

PART- B

(5 X 10 =50 Marks)

2. a) Discuss friend functions in C++. [5M]L2
- b) Illustrate constructor and destructor with a suitable program. [5M]L2

OR

3. a) Explain the features of C++. [5M]L2
- b) Demonstrate operator overloading with an example program. [5M]L3
4. a) What is inheritance? Explain the different types of inheritance with the help of an example. [7M]L2
- b) List the applications of Stack. [3M]L1

OR

5. a) Write a c++ program to catch all exceptions. [5M]L3
- b) What is singly linked list? How to represent it? Discuss. [5M]L2
6. a) Create a binary search tree for the following numbers start from an empty [5M]L3
binary search tree. 45,26,10,60,70,30,40.
- b) Why AVL tree is called Balanced tree? Explain. [5M]L2

OR

7. a) Demonstrate Tree travels with a suitable example. [5M]L3
b) Explain how to represent Binary trees using linked list. [5M]L2

8. a) Explain Binary search with suitable example. [5M]L3
b) Explain heap sort. Construct heap sort for the initial key set 42, 23, 74, 11, 65, 58, 94, 36, 99, 87. [5M]L3

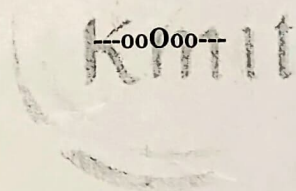
OR

9. a) What is Priority queues? List applications of priority queues. [4M]L1
b) Write a C++ program to perform merge sort. [6M]L3

10. a) Explain the method of representing graphs by using matrices? [5M]L2
b) Explain Boyer-moor algorithm. [5M]L2

OR

11. a) What are the different graph traversals? Explain DFS with a suitable example. [5M]L2
b) What is meant by string matching? List the applications of string matching. [5M]L2



Code No: 213AC

KR21



KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

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Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year I Semester Supplementary Examinations, August / September- 2023

Data Structures through C++

(Common to CSE, IT, CSE (AI&ML), CSE (DATA SCIENCE))

Time: 3 hours

Max. Marks: 70

Note: This question paper contains two parts A and B.

Part A is compulsory which carries 20 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(2 X 10 =20 Marks)

1. a) What is an array? Explain array types [2M]
- b) Define Constructor and Destructor [2M]
- c) List out applications of stack [2M]
- d) What is meant by inheritance and polymorphism [2M]
- e) Define height of a binary tree.. [2M]
- f) What is hash function? [2M]
- g) Compare quick sort and merge sort [2M]
- h) Define a max heap. [2M]
- i) What is directed graph? [2M]
- j) What are the applications of graphs? [2M]

PART- B

(5 X 10 =50 Marks)

2. a) What is Constructor? Explain various types of constructors with an examples. [6M]
- b) Explain about function overloading with example [4M]

OR

3. a) Write a C++ program to overload + operator to concatenate two strings. [6M]
- b) Explain about different types of access specifiers with example. [4M]

4. a) Explain about throwing an exception [5M]
- b) Explain array representation of stack [5M]

OR

5. a) Discuss in detail about queue with example [6M]
- b) Briefly explain about applications of stack. [4M]

P.T.O

6. a) What is tree traversal? Explain in detail various tree traversals techniques. [5M]

b) Explain the following: a) AVL tree properties b) Threaded binary tree [5M]

OR

7. a) What are the properties of a binary tree? [3M]

b) Create binary search tree for the following elements (23, 32, 24, 36, 15, 12, 39, 2, 19).

Discuss about the height of the above binary search tree. [7M]

8. a) Explain about Insertion and deletion operations in max heap [5M]

b) Write C++ program for heap sort technique. [5M]

OR

9. a) Write C++ program for Quick sort technique [5M]

b) Write C++ program for Merge sort technique [5M]

10. a) Explain the following: a) Adjacency matrix b) Minimum cost spanning trees [5M]

b) Describe about string matching algorithms [5M]

OR

11. a) Write an algorithm to traverse a graph using breadth first search. [5M]

b) Explain about the Depth-First-search method. [5M]

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Code No:213AC

KR21

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year I Semester Regular Examinations, March- 2023

Data Structures through C++

(Common to IT, CSE (AI&ML), CSE (DATA SCIENCE))

Time: 3 hours

Max. Marks: 70

Note: This question paper contains two parts A and B.

Part A is compulsory which carries 20 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b as sub questions.

<i>Q.No</i>	PART -A	Marks
1. (a)	Justify which is the most important feature of C++, and why?	2
(b)	Write the sequence of constructor & destructor execution in inheritance in C++ program.	2
(c)	Explain the advantages of inheritance with suitable example.	2
(d)	Which data structure is used during function calls, and how they are used?	2
(e)	Illustrate the binary tree representation through arrays.	2
(f)	Define binary search tree in a recursive form. List its properties.	2
(g)	Why is hashing considered as best data structure, justify?	2
(h)	What is priority queue? Differentiate between the max heap and min heap.	2
(i)	Briefly explain how graphs are different from trees.	2
(j)	What is the need of string matching algorithms?	2

PART- B

Q.2(a)	What is the use of friend function? Explain with a suitable program?	5
(b)	Write a program to swap two numbers using references, function templates.	5
OR		
Q.3(a)	Explain all the different access specifiers in C++ with suitable example.	5
(b)	Why references are felt needed in C++ when pointers already exist in C?	5
Q.4(a)	Implement stack operations using linked list, Also give the time complexity of each operation.	5
(b)	Explain all different types of inheritance in C++.	5

P.T.O

OR

Q.5(a) Explain all asymptotic notations in measuring algorithm time complexity. 5

(b) Write C++ program to implement circular single linked list. Use exception handling where ever needed. 5

Q.6(a) Write and explain the algorithm for deleting an element in the BST in all cases. 5

(b) Which Binary search traversal is sufficient to construct the binary search tree? Explain the steps in the process of obtain the tree. 5

OR

Q.7(a) Explain how a binary threaded tree is different from binary search tree. Justify in detail which is better and how is it better. 5

(b) List different height balanced trees. Write the mechanism how height balance is achieved with suitable examples. 5

Q.8(a) What are priority queues? Implement the heap sort and give its time and space complexity. 5

(b) Compare the working and time complexity of binary and linear search. 5

OR

Q.9(a) Explain the Quick sort algorithm for the given array elements 2, 38, 76, 64, 18, 32, 96, 4, 5, 12 with random pivot element. Derive the time complexity. 5

(b) Write a C++ program to implement radix sort. Also find its time and space complexity. 5

Q.10(a) What is the significance of the minimum spanning tree? Write a program for prim's minimum spanning tree. 5

(b) Write and explain Boyer-Moore Pattern matching algorithm. 5

OR

Q.11(a) Write and explain Knuth-Morris-Pratt pattern matching algorithm. 5

(b) Explain the Graph traversal methods with suitable examples. 5

END OF THE QUESTION PAPER

Code No:213AC

KR2-1

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year I Semester Regular Examinations, March- 2023

Data Structures through C++
(Computer Science and Engineering)



Time: 3 hours

Max. Marks: 70

Note: This question paper contains two parts A and B.

Part A is compulsory which carries 20 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b as sub questions.

Q.No	PART – A	Marks
1. (a)	Define Data Abstraction?	2
(b)	How templates are useful in C++	2
(c)	Convert the infix expression $A+B*C*D/E+F-G*H$ into prefix expression.	2
(d)	What is the difference in adding new item at the head, non-head node in linked list.	2
(e)	Why Big O notation has more significance compared to other notations?	2
(f)	Differentiate between complete binary tree and full binary tree?	2
(g)	How comparison sort is different from counting sort?	2
(h)	What is rehashing?	2
(i)	Define spanning tree.	2
(j)	What are the applications of graph?	2
	PART- B	
Q.2(a)	Explain all the OOPS Concepts.	5
(b)	Write a C++ program to find the sum of the array using template functions.	5
	OR	
Q.3(a)	Differentiate between function overloading and function overriding.	5
(b)	Explain with a suitable example the concept of copy constructor.	5
Q.4(a)	Implement in C++ doubly linked list. Also give the time complexity of each operation.	5
(b)	Write a C++ Program to implement circular queue. Give the complexity of the operations.	5

OR

P.T.O

- Q.5(a)** Compare and contrast stack implementation using arrays and linked list. 5
- (b) Write an algorithm to evaluate arithmetic expression using stack? 5
- Q.6(a)** Write a C++ Program and explain to perform common operations on binary search tree? 5
- (b) To construct a binary search tree which binary tree traversal are sufficient? Justify your answer by taking a suitable example? 5
- OR**
- Q.7(a)** Explain the array representation of binary trees with a suitable example. 5
- (b) Define tree? Explain all the terms related to trees 5
- Q.8(a)** Create a max heap for the given data 20,45,65,34,24,78,90,5. Why arrays are used and not linked structures in creating heap 5
- (b) What is collision in hashing? Explain different mechanism to deal collision? 5
- OR**
- Q.9(a)** Compare different sorting techniques? In what scenario merge sort is good? 5
- (b) Write a C++ program to implement counting sort? Also gives its time and space complexity 5
- Q.10(a)** Write a program for Kruskal's minimum spanning tree? What is the significance of the minimum spanning tree? 5
- (b) Explain the DFS and BFS traversal methods in graph with an example. 5
- OR**
- Q.11(a)** Write and explain Knuth-Morris-Pratt pattern matching algorithm? 5
- (b) Trace the Boyer-Moore algorithm for the text="ABACCABBACCAA" and the pattern="ACC" 5

END OF THE QUESTION PAPER

**KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY**

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Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year I Semester Regular Examinations, February-2024

**Data Structures through C++
(COMPUTER SCIENCE AND ENGINEERING)****Time: 3 hours****Max. Marks: 70****Note:** This question paper contains two parts A and B.

Part A is compulsory which carries 20 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b, c as sub questions.

PART- A**(2 X 10 =20 Marks)**

1. a) Explain the structure of C++ Program with an example. [2M]L2
- b) Write the structure of a class with an example. [2M]L2
- c) What is recursion? [2M]L1
- d) Why do we need Data Structure. [2M]L1
- e) What are the operations of the stack? [2M]L1
- f) List the applications of queue. [2M]L1
- g) Define a heap. How can it be used to represent a priority queue [2M]L1
- h) Define Complete binary tree. [2M]L1
- i) Name the different ways of representing a graph. [2M]L1
- j) List the problems in hashing. [2M]L1

PART- B**(5 X 10 =50 Marks)**

2. What is a Constructor? Explain different types of constructors with example programs. [10M]L3

OR

3. a) Write a program to find the areas of different geometrical shapes using Parameterized constructor. [5M]L3
- b) Write a C++ program to find the sum of the array using template functions. [5M]L3

4. Write a C++ program to implement the operations of singly linked list. [10M]L3

OR

5. Explain the addition of polynomial with a program. [10M]L3
6. Write an algorithm to evaluate Postfix expression with an example. [10M]L3

OR

7. a) Differentiate Queues with circular queue. [5M]L3
- b) Write a program to perform basic operations on a queue. [5M]L3

8. Explain Priority queue with an example. What are the applications of priority Queues.

[10M]L3

OR

9. To construct a binary search tree which binary tree traversal are sufficient?

Justify your answer by taking a suitable example.

[10M]L3

10. Differentiate between linear and Binary Search. Write a program to implement Binary Search.

[10M]L3

OR

11. Explain the DFS and BFS traversal methods in graph with an example.

[10M]L3

---ooOoo---

KIT

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year I Semester Regular Examinations, February-2024

**Data Structures through C++**

(Common to IT, CSE (AI&ML), CSE (DATA SCIENCE))

Time: 3 hours

Max. Marks: 70

Note: This question paper contains two parts A and B.

Part A is compulsory which carries 20 marks. Answer all questions in Part A.

Part B consists of 5 Units. Answer any one full question from each unit.

Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(2 X 10 =20 Marks)

1. a) Write about destructor with an example. [2M]L2
- b) What are different types of parameter passing techniques? [2M]L1
- c) What is column order majoring [2M]L1
- d) Differentiate direct and indirect recursion [2M]L2
- e) What are the operations of the queue? [2M]L1
- f) List the applications of stack [2M]L1
- g) Define binary tree and give binary tree node structure [2M]L1
- h) What happens to the structure of a max heap when removing the maximum element? [2M]L1
- i) What is binary search? [2M]L1
- j) What is Extensible Hashing [2M]L1

PART- B

(5 X 10 =50 Marks)

2. Explain about generic programming with an example program [10M]L3

OR

3. a) Define a class and object. Explain different types of access specifiers [5M]L3
- b) Explain about default arguments with example program. [5M]L3
4. With neat diagrams, explain the Insert and Delete operations in Doubly Linked List data structure [10M]L3

OR

5. Define a sparse matrix. Write the working of sparse matrix using array Implementation. [10M]L3
6. a) Convert the infix expression $a/b-c+d*e-a*c$ into postfix expression using a stack [5M]L3
- b) What is LIFO? How to represent a queue? Explain. [5M]L3

OR

7. a) Explain the real time applications of queue. [5M]L3
- b) Write a program to perform basic operations on a stack. [5M]L3

8. Write an algorithm for in-order traversal of a binary tree. Explain with an example.

[10M]L3

OR

9. Write an algorithm to construct a Binary Search Tree from a given list of elements.

Illustrate the same with 34,13,58,29,47,6,71,62,88,25

[10M]L3

10.a) Write a C++ program to perform Breadth First Search of a graph

[5M]L3

b) What are various representations of graphs. Explain with suitable example.

[5M]L3

OR

11. Compare and contrast BFS with DFS with an example

[10M]L3

---ooOoo---

KMII